

Functions of Two or Three Real Variables (2005–2026 Archive)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

78

TOTAL MARKS

152 M

TEST DURATION

234 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2005–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	44	+1 / +2 Marks	$-1/3$ / $-2/3$ Marks	4 Options (1 Correct)
Multiple Select (MSQ)	13	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	21	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty ($1/3$ of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q52 • ID: JAM_2026_Q52)

NAT

+2 M

Neg: Nil (0)



Q.52

Let the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 4x^2 \tan^{-1}\left(\frac{y}{2x}\right) + y^3 \tan^{-1}\left(\frac{x}{4y^2}\right), & xy \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

Then $\left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}\right)\right)$ at $(0,0)$ is _____ (rounded off to two decimal places).

Question 2 (JAM 2026 • Q46 • ID: JAM_2026_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46

Let $z = \cos(4x + 5y)$, where $x = \frac{\pi}{2} + 2\theta$, $y = -\left(\frac{\pi}{4} + \theta\right)$.Then $\left.\frac{dz}{d\theta}\right|_{\theta=\frac{\pi}{4}} = \text{_____}$ (rounded off to one decimal place).

Question 3 (JAM 2026 • Q36 • ID: JAM_2026_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36

$$\text{Let } f(x, y) = \begin{cases} \frac{x^2 y}{1+x^2} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Which of the following statements is/are TRUE?

(A) $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exists.(B) $\frac{\partial f}{\partial x}$ is continuous at the point $(0, 1)$.(C) $\frac{\partial f}{\partial y}$ is continuous at the point $(0, 1)$.(D) $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y)}{\sqrt{x^2+y^2}}$ does not exist.

Question 4 (JAM 2026 • Q17 • ID: JAM_2026_Q17)

MCQ

+2 M

Neg: -0.67

Q.17	Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by $f(x, y) = \begin{cases} \frac{5x^2y - 4y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ Which ONE of the following statements is FALSE?
(A)	f is continuous at $(0, 0)$
(B)	$\frac{\partial f}{\partial x}$ at $(0, 1) = 0$
(C)	$\frac{\partial f}{\partial y}$ at $(1, 0) = 5$
(D)	$\frac{\partial f}{\partial y}$ is continuous at $(0, 0)$

Question 5 (JAM 2026 • Q15 • ID: JAM_2026_Q15)

MCQ

+2 M

Neg: -0.67

Q.15	Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2 \text{ for all } (x, y) \in \mathbb{R}^2.$ Which ONE of the following statements is TRUE?
(A)	f has local maxima at exactly two points.
(B)	f has local minima at exactly two points.
(C)	f has local minima at exactly three points.
(D)	f has exactly four saddle points.

Question 6 (JAM 2025 • Q50 • ID: JAM_2025_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{(x^2 - y^2)xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, the value of $\frac{\partial f}{\partial y}(1, 0) - \frac{\partial f}{\partial x}(0, 2)$ is equal to _____ (rounded off to two decimal places).

Question 7 (JAM 2025 • Q34 • ID: JAM_2025_Q34)

MSQ

+2 M

Neg: Nil (0)

Q.34 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{(x^2 + \sin x)y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, which of the following is/are TRUE?

- (A) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists and $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$
- (B) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists and $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$
- (C) f is differentiable at $(0, 0)$
- (D) f is NOT differentiable at $(0, 0)$

Question 8 (JAM 2025 • Q31 • ID: JAM_2025_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy^2 + y^5}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Then, which of the following is/are TRUE?

- (A) The iterated limits $\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right)$ and $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right)$ exist
- (B) Exactly one of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists at $(0, 0)$
- (C) Both the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at $(0, 0)$
- (D) f is NOT differentiable at $(0, 0)$

Question 9 (JAM 2025 • Q27 • ID: JAM_2025_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = e^y(x^2 + y^2) \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Then, which one of the following is TRUE?

- (A) The number of points at which f has a local minimum is 2
- (B) The number of points at which f has a local maximum is 2
- (C) The number of points at which f has a local minimum is 1
- (D) The number of points at which f has a local maximum is 1

Question 10 (JAM 2024 • Q35 • ID: JAM_2024_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35

Let

$$S = \{(x, y) \in \mathbb{R}^2: x > 0, y > 0\},$$

and $f: S \rightarrow \mathbb{R}$ be given by

$$f(x, y) = 2x^2 + 3y^2 - \log x - \frac{1}{6} \log y.$$

Then, which of the following statements is/are TRUE?

- (A) There is a unique point in S at which $f(x, y)$ attains a local maximum
- (B) There is a unique point in S at which $f(x, y)$ attains a local minimum
- (C) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is bounded
- (D) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is unbounded

Question 11 (JAM 2024 • Q30 • ID: JAM_2024_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Define the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = 12xy e^{-(2x+3y-2)}.$$

If (a, b) is the point of local maximum of f , then $f(a, b)$ equals

(A) 2

(B) 6

(C) 12

(D) 0

Question 12 (JAM 2024 • Q2 • ID: JAM_2024_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 For a twice continuously differentiable function $g: \mathbb{R} \rightarrow \mathbb{R}$, define

$$u_g(x, y) = \frac{1}{y} \int_{-y}^y g(x+t) dt \quad \text{for } (x, y) \in \mathbb{R}^2, \quad y > 0.$$

Which one of the following holds for all such g ?

(A) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$

(B) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$

(C) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$

(D) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$

Question 13 (JAM 2023 • Q51 • ID: JAM_2023_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined as $f(x, y, z) = x^3 + y^3 + z^3$, and let $L: \mathbb{R}^3 \rightarrow \mathbb{R}$ be the linear map satisfying

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{f(1+x, 1+y, 1+z) - f(1,1,1) - L(x,y,z)}{\sqrt{x^2 + y^2 + z^2}} = 0.$$

Then $L(1, 2, 4)$ is equal to _____. (rounded off to two decimal places)

Question 14 (JAM 2023 • Q41 • ID: JAM_2023_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined as follows:

$$f(x, y) = \begin{cases} (x^2 - 1)^2 \cos^2 \left(\frac{y^2}{(x^2 - 1)^2} \right) & \text{if } x \neq \pm 1 \\ 0 & \text{if } x = \pm 1. \end{cases}$$

The number of points of discontinuity of $f(x, y)$ is equal to _____.

Question 15 (JAM 2023 • Q37 • ID: JAM_2023_Q37)

MSQ

+2 M

Neg: Nil (0)

Q. 37 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as follows:

$$f(x, y) = \begin{cases} \frac{x^4 y^3}{x^6 + y^6} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then

(A) $\lim_{t \rightarrow 0} \frac{f(t, t) - f(0, 0)}{t}$ exists and equals $\frac{7}{2}$

(B) $\left. \frac{\partial f}{\partial x} \right|_{(0,0)}$ exists and equals 0

(C) $\left. \frac{\partial f}{\partial y} \right|_{(0,0)}$ exists and equals 0

(D) $\lim_{t \rightarrow 0} \frac{f(t, 2t) - f(0, 0)}{t}$ exists and equals $\frac{1}{3}$

Question 16 (JAM 2023 • Q36 • ID: JAM_2023_Q36)

MSQ

+2 M

Neg: Nil (0)

Q. 36 A subset $S \subseteq \mathbb{R}^2$ is said to be *bounded* if there is an $M > 0$ such that $|x| \leq M$ and $|y| \leq M$ for all $(x, y) \in S$. Which of the following subsets of $\mathbb{R}^2$ is/are bounded?

(A) $\{(x, y) \in \mathbb{R}^2 : e^{x^2} + y^2 \leq 4\}$

(B) $\{(x, y) \in \mathbb{R}^2 : x^4 + y^2 \leq 4\}$

(C) $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4\}$

(D) $\{(x, y) \in \mathbb{R}^2 : e^{x^3} + y^2 \leq 4\}$

Question 17 (JAM 2023 • Q17 • ID: JAM_2023_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f(x, y) = \ln(1 + x^2 + y^2)$ for $(x, y) \in \mathbb{R}^2$. Define

$$P = \frac{\partial^2 f}{\partial x^2} \Big|_{(0,0)} \quad Q = \frac{\partial^2 f}{\partial x \partial y} \Big|_{(0,0)}$$

$$R = \frac{\partial^2 f}{\partial y \partial x} \Big|_{(0,0)} \quad S = \frac{\partial^2 f}{\partial y^2} \Big|_{(0,0)}$$

Then

(A) $PS - QR > 0$ and $P < 0$

(B) $PS - QR > 0$ and $P > 0$

(C) $PS - QR < 0$ and $P > 0$

(D) $PS - QR < 0$ and $P < 0$

Question 18 (JAM 2023 • Q16 • ID: JAM_2023_Q16)

MCQ

+2 M

Neg: -0.67

Q. 16 Let $f(x, y) = e^{x^2+y^2}$ for $(x, y) \in \mathbb{R}^2$, and a_n be the determinant of the matrix

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

evaluated at the point $(\cos(n), \sin(n))$. Then the limit $\lim_{n \rightarrow \infty} a_n$ is

- (A) non-existent
- (B) 0
- (C) $6e^2$
- (D) $12e^2$

Question 19 (JAM 2023 • Q8 • ID: JAM_2023_Q8)

MCQ

+1 M

Neg: -0.33

Q. 8 Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (e^x \cos(y), e^x \sin(y))$. Then the number of points in $\mathbb{R}^2$ that do NOT lie in the range of f is

- (A) 0
- (B) 1
- (C) 2
- (D) infinite

Question 20 (JAM 2022 • Q56 • ID: JAM_2022_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56

Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = x^2 + 2y^2 - x \quad \text{for } (x, y) \in \mathbb{R}^2.$$

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ and $E = \{(x, y) \in \mathbb{R}^2 : \frac{x^2}{4} + \frac{y^2}{9} \leq 1\}$.

Consider the sets

$$D_{\max} = \{(a, b) \in D : f \text{ has absolute maximum on } D \text{ at } (a, b)\},$$

$$D_{\min} = \{(a, b) \in D : f \text{ has absolute minimum on } D \text{ at } (a, b)\},$$

$$E_{\max} = \{(c, d) \in E : f \text{ has absolute maximum on } E \text{ at } (c, d)\},$$

$$E_{\min} = \{(c, d) \in E : f \text{ has absolute minimum on } E \text{ at } (c, d)\}.$$

Then the total number of elements in the set

$$D_{\max} \cup D_{\min} \cup E_{\max} \cup E_{\min}$$

is equal to _____.

Question 21 (JAM 2022 • Q51 • ID: JAM_2022_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t .

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$. Let $f: D \rightarrow \mathbb{R}$ and $g: D \rightarrow \mathbb{R}$ be defined by $f(0, 0) = g(0, 0) = 0$ and

$$f(x, y) = [x^2 + y^2] \frac{x^2 y^2}{x^4 + y^4}, \quad g(x, y) = [y^2] \frac{xy}{x^2 + y^2}$$

for $(x, y) \neq (0, 0)$. Let E be the set of points of D at which both f and g are discontinuous. The number of elements in the set E is _____.

Question 22 (JAM 2022 • Q42 • ID: JAM_2022_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42

Consider the function $u: \mathbb{R}^3 \rightarrow \mathbb{R}$ given by

$$u(x_1, x_2, x_3) = x_1 x_2^4 x_3^2 - x_1^3 x_3^4 - 26x_1^2 x_2^2 x_3^3.$$

Let $c \in \mathbb{R}$ and $k \in \mathbb{N}$ be such that

$$x_1 \frac{\partial u}{\partial x_2} + 2x_2 \frac{\partial u}{\partial x_3}$$

evaluated at the point (t, t^2, t^3) , equals ct^k for every $t \in \mathbb{R}$. Then the value of k is equal to _____.

Question 23 (JAM 2022 • Q36 • ID: JAM_2022_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36

Let M be a positive real number and let $u, v: \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous functions satisfying

$$\sqrt{u(x, y)^2 + v(x, y)^2} \geq M\sqrt{x^2 + y^2} \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

$$F(x, y) = (u(x, y), v(x, y)) \quad \text{for } (x, y) \in \mathbb{R}^2.$$

Then which of the following is/are true?

(A)

 F is injective.

(B)

If K is open in $\mathbb{R}^2$, then $F(K)$ is open in $\mathbb{R}^2$.

(C)

If K is closed in $\mathbb{R}^2$, then $F(K)$ is closed in $\mathbb{R}^2$.

(D)

If E is closed and bounded in $\mathbb{R}^2$, then $F^{-1}(E)$ is closed and bounded in $\mathbb{R}^2$.

Question 24 (JAM 2022 • Q15 • ID: JAM_2022_Q15)

MCQ

+2 M

Neg: -0.67

Q.15

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t . Define functions $h: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x, y) = \begin{cases} \frac{-1}{x^2 - y} & \text{if } x^2 \neq y, \\ 0 & \text{if } x^2 = y \end{cases} \quad \text{and} \quad g(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Then which one of the following is FALSE?

(A)

$$\lim_{(x, y) \rightarrow (\sqrt{2}, \pi)} \cos\left(\frac{x^2 y}{x^2 + 1}\right) = \frac{-1}{2}.$$

(B)

$$\lim_{(x, y) \rightarrow (\sqrt{2}, 2)} e^{h(x, y)} = 0.$$

(C)

$$\lim_{(x, y) \rightarrow (e, e)} \ln(x^{y - [y]}) = e - 2.$$

(D)

$$\lim_{(x, y) \rightarrow (0, 0)} e^{2y} g(x) = 1.$$

Question 25 (JAM 2022 • Q10 • ID: JAM_2022_Q10)

MCQ

+1 M

Neg: -0.33

Q.10

Let $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Consider the functions

$$u : \mathbb{R}^2 - \{(0,0)\} \rightarrow \mathbb{R} \quad \text{and} \quad v : \mathbb{R}^2 - \{(0,0)\} \rightarrow \mathbb{R}$$

given by

$$u(x,y) = x - \frac{x}{x^2 + y^2} \quad \text{and} \quad v(x,y) = y + \frac{y}{x^2 + y^2}.$$

The value of the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ at the point $(\cos \theta, \sin \theta)$ is equal to

(A) $4 \sin \theta$.(B) $4 \cos \theta$.(C) $4 \sin^2 \theta$.(D) $4 \cos^2 \theta$.

Question 26 (JAM 2022 • Q9 • ID: JAM_2022_Q9)

MCQ

+1 M

Neg: -0.33

Q.9

Let c be a positive real number and let $u: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$u(x, t) = \frac{1}{2c} \int_{x-ct}^{x+ct} e^{s^2} ds \quad \text{for } (x, t) \in \mathbb{R}^2.$$

Then which one of the following is true?

(A) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{on } \mathbb{R}^2.$

(B) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{on } \mathbb{R}^2.$

(C) $\frac{\partial u}{\partial t} \frac{\partial u}{\partial x} = 0 \quad \text{on } \mathbb{R}^2.$

(D) $\frac{\partial^2 u}{\partial t \partial x} = 0 \quad \text{on } \mathbb{R}^2.$

Question 27 (JAM 2021 • Q34 • ID: JAM_2021_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 34 Consider the two functions $f(x, y) = x + y$ and $g(x, y) = xy - 16$ defined on $\mathbb{R}^2$. Then

- (A) the function f has no global extreme value subject to the condition $g = 0$.
- (B) the function f attains global extreme values at $(4, 4)$ and $(-4, -4)$ subject to the condition $g = 0$.
- (C) the function g has no global extreme value subject to the condition $f = 0$.
- (D) the function g has a global extreme value at $(0, 0)$ subject to the condition $f = 0$.

Question 28 (JAM 2021 • Q33 • ID: JAM_2021_Q33)

MSQ

+2 M

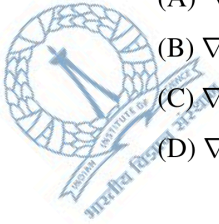
Neg: Nil (0)

Q. 33 Let $D = \mathbb{R}^2 \setminus \{(0, 0)\}$. Consider the two functions $u, v : D \rightarrow \mathbb{R}$ defined by

$$u(x, y) = x^2 - y^2 \text{ and } v(x, y) = xy.$$

Consider the gradients ∇u and ∇v of the functions u and v , respectively. Then

- (A) ∇u and ∇v are parallel at each point (x, y) of D .
 (B) ∇u and ∇v are perpendicular at each point (x, y) of D .
 (C) ∇u and ∇v do not exist at some points (x, y) of D .
 (D) ∇u and ∇v at each point (x, y) of D span $\mathbb{R}^2$.



MA

11 / 17

Question 29 (JAM 2021 • Q23 • ID: JAM_2021_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let $D \subseteq \mathbb{R}^2$ be defined by $D = \mathbb{R}^2 \setminus \{(x, 0) : x \in \mathbb{R}\}$. Consider the function $f : D \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x \sin \frac{1}{y}.$$

Then

- (A) f is a discontinuous function on D .
 (B) f is a continuous function on D and cannot be extended continuously to any point outside D .
 (C) f is a continuous function on D and can be extended continuously to $D \cup \{(0, 0)\}$.
 (D) f is a continuous function on D and can be extended continuously to the whole of $\mathbb{R}^2$.

Question 30 (JAM 2020 • Q53 • ID: JAM_2020_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 The minimum value of the function $f(x, y) = x^2 + xy + y^2 - 3x - 6y + 11$ is _____.

Question 31 (JAM 2020 • Q45 • ID: JAM_2020_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 Let $f(x, y) = e^x \sin y$, $x = t^3 + 1$ and $y = t^4 + t$. Then $\frac{df}{dt}$ at $t = 0$ is _____. (rounded off to two decimal places)

MA

14 / 17

Question 32 (JAM 2020 • Q17 • ID: JAM_2020_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f(x, y, z) = x^3 + y^3 + z^3 - 3xyz$. A point at which the gradient of the function f is equal to zero is

- (A) $(-1, 1, -1)$ (B) $(-1, -1, -1)$ (C) $(-1, 1, 1)$ (D) $(1, -1, 1)$

Question 33 (JAM 2020 • Q15 • ID: JAM_2020_Q15)

MCQ

+2 M

Neg: -0.67

Q. 15 Let $f(x, y) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0, y = 0 \\ y^2 \sin \frac{1}{y}, & y \neq 0, x = 0 \\ 0, & x = y = 0. \end{cases}$

Which of the following is true at $(0, 0)$?

- (A) f is not continuous
 (B) $\frac{\partial f}{\partial x}$ is continuous but $\frac{\partial f}{\partial y}$ is not continuous
 (C) f is not differentiable
 (D) f is differentiable but both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not continuous

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Question 34 (JAM 2020 • Q6 • ID: JAM_2020_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 If the directional derivative of the function $z = y^2 e^{2x}$ at $(2, -1)$ along the unit vector $\vec{b} = \alpha \hat{i} + \beta \hat{j}$ is zero, then $|\alpha + \beta|$ equals

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$

(D) $2\sqrt{2}$

Question 35 (JAM 2020 • Q5 • ID: JAM_2020_Q5)

MCQ

+1 M

Neg: -0.33

JAM 2020

MATHEMATICS - MA

Q. 5 If the equation of the tangent plane to the surface $z = 16 - x^2 - y^2$ at the point $P(1, 3, 6)$ is $ax + by + cz + d = 0$, then the value of $|d|$ is

(A) 16

(B) 26

(C) 36

(D) 46

Question 36 (JAM 2020 • Q4 • ID: JAM_2020_Q4)

MCQ

+1 M

Neg: -0.33

Q. 4 Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $f(x, y) = g(y) + xg'(y)$, then

(A) $\frac{\partial f}{\partial x} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(B) $\frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

(C) $\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial y}$

(D) $\frac{\partial f}{\partial y} + x \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x}$

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Question 37 (JAM 2019 • Q54 • ID: JAM_2019_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 The number of critical points of the function

$$f(x, y) = (x^2 + 3y^2)e^{-(x^2+y^2)}$$

is _____

Question 38 (JAM 2019 • Q48 • ID: JAM_2019_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let

$$f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 - y^2}, & x^2 - y^2 \neq 0 \\ 0, & x^2 - y^2 = 0. \end{cases}$$

Then the directional derivative of f at $(0, 0)$ in the direction of $\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$ is _____

Question 39 (JAM 2019 • Q39 • ID: JAM_2019_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let

$$f(x, y) = \begin{cases} \frac{|x|}{|x| + |y|} \sqrt{x^4 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

Then at $(0, 0)$,

- (A) f is continuous
- (B) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y}$ does not exist
- (C) $\frac{\partial f}{\partial x}$ does not exist and $\frac{\partial f}{\partial y} = 0$
- (D) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

Question 40 (JAM 2019 • Q29 • ID: JAM_2019_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 For $\beta \in \mathbb{R}$, define

$$f(x, y) = \begin{cases} \frac{x^2 |x|^\beta y}{x^4 + y^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then, at $(0, 0)$, the function f is

- (A) continuous for $\beta = 0$
- (B) continuous for $\beta > 0$
- (C) not differentiable for any β
- (D) continuous for $\beta < 0$

Question 41 (JAM 2019 • Q26 • ID: JAM_2019_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The tangent line to the curve of intersection of the surface $x^2 + y^2 - z = 0$ and the plane $x + z = 3$ at the point $(1, 1, 2)$ passes through

- (A) $(-1, -2, 4)$
- (B) $(-1, 4, 4)$
- (C) $(3, 4, 4)$
- (D) $(-1, 4, 0)$

Question 42 (JAM 2019 • Q22 • ID: JAM_2019_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 The function

$$f(x, y) = x^3 + 2xy + y^3$$

has a saddle point at

- (A) $(0, 0)$
- (B) $\left(-\frac{2}{3}, -\frac{2}{3}\right)$
- (C) $\left(-\frac{3}{2}, -\frac{3}{2}\right)$
- (D) $(-1, -1)$

Question 43 (JAM 2019 • Q8 • ID: JAM_2019_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 The equation of the tangent plane to the surface $x^2 z + \sqrt{8 - x^2 - y^4} = 6$ at the point $(2, 0, 1)$ is

- (A) $2x + z = 5$
- (B) $3x + 4z = 10$
- (C) $3x - z = 10$
- (D) $7x - 4z = 10$

Question 44 (JAM 2019 • Q3 • ID: JAM_2019_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. Define $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) = g(x^2 + y^2 - 2z^2).$$

Then $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

- (A) $4(x^2 + y^2 - 4z^2) g''(x^2 + y^2 - 2z^2)$
- (B) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2)$
- (C) $4(x^2 + y^2 - 2z^2) g''(x^2 + y^2 - 2z^2)$
- (D) $4(x^2 + y^2 + 4z^2) g''(x^2 + y^2 - 2z^2) + 8g'(x^2 + y^2 - 2z^2)$

Question 45 (JAM 2018 • Q53 • ID: JAM_2018_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Suppose x, y, z are positive real numbers such that $x + 2y + 3z = 1$. If M is the maximum value of xyz^2 , then the value of $\frac{1}{M}$ is _____

MA

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Question 46 (JAM 2018 • Q45 • ID: JAM_2018_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 Let $f(x, y) = \sqrt{x^3 y} \sin\left(\frac{\pi}{2} e^{\left(\frac{y}{x}-1\right)}\right) + xy \cos\left(\frac{\pi}{3} e^{\left(\frac{x}{y}-1\right)}\right)$ for $(x, y) \in \mathbb{R}^2, x > 0, y > 0$.

Then $f_x(1, 1) + f_y(1, 1) =$ _____

Question 47 (JAM 2018 • Q44 • ID: JAM_2018_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{x^2 y (x - y)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Then $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ at the point $(0, 0)$ is _____

Question 48 (JAM 2018 • Q42 • ID: JAM_2018_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let $\phi(x, y, z) = 3y^2 + 3yz$ for $(x, y, z) \in \mathbb{R}^3$. Then the absolute value of the directional derivative of ϕ in the direction of the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z}{-2}$, at the point $(1, -2, 1)$ is _____

Question 49 (JAM 2018 • Q18 • ID: JAM_2018_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 Let $a, b \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. If $z = e^u f(v)$, where $u = ax + by$ and $v = ax - by$, then which one of the following is TRUE?

(A) $b^2 z_{xx} - a^2 z_{yy} = 4a^2 b^2 e^u f'(v)$

(B) $b^2 z_{xx} - a^2 z_{yy} = -4e^u f'(v)$

(C) $bz_x + az_y = abz$

(D) $bz_x + az_y = -abz$

Question 50 (JAM 2018 • Q17 • ID: JAM_2018_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f(x, y) = \begin{cases} \frac{xy}{(x^2+y^2)^\alpha}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

Then which one of the following is TRUE for f at the point $(0, 0)$?

- (A) For $\alpha = 1$, f is continuous but not differentiable
- (B) For $\alpha = \frac{1}{2}$, f is continuous and differentiable
- (C) For $\alpha = \frac{1}{4}$, f is continuous and differentiable
- (D) For $\alpha = \frac{3}{4}$, f is neither continuous nor differentiable

Question 51 (JAM 2018 • Q6 • ID: JAM_2018_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 In $\mathbb{R}^3$, the cosine of the acute angle between the surfaces $x^2 + y^2 + z^2 - 9 = 0$ and $z - x^2 - y^2 + 3 = 0$ at the point $(2, 1, 2)$ is

- (A) $\frac{8}{5\sqrt{21}}$
- (B) $\frac{10}{5\sqrt{21}}$
- (C) $\frac{8}{3\sqrt{21}}$
- (D) $\frac{10}{3\sqrt{21}}$

Question 52 (JAM 2018 • Q5 • ID: JAM_2018_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 The tangent plane to the surface $z = \sqrt{x^2 + 3y^2}$ at $(1, 1, 2)$ is given by

- (A) $x - 3y + z = 0$
- (B) $x + 3y - 2z = 0$
- (C) $2x + 4y - 3z = 0$
- (D) $3x - 7y + 2z = 0$

MA

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Question 53 (JAM 2017 • Q44 • ID: JAM_2017_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let P be the point on the surface $z = \sqrt{x^2 + y^2}$ closest to the point $(4, 2, 0)$. Then the square of the distance between the origin and P is _____

Question 54 (JAM 2017 • Q33 • ID: JAM_2017_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. Then which of the following statements is/are TRUE?

- (A) If f is differentiable at $(0,0)$, then all directional derivatives of f exist at $(0,0)$
- (B) If all directional derivatives of f exist at $(0,0)$, then f is differentiable at $(0,0)$
- (C) If all directional derivatives of f exist at $(0,0)$, then f is continuous at $(0,0)$
- (D) If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous in a disc centered at $(0,0)$, then f is differentiable at $(0,0)$

Question 55 (JAM 2017 • Q23 • ID: JAM_2017_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $f(x, y) = \frac{x^2 y}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Then

- (A) $\frac{\partial f}{\partial x}$ and f are bounded
- (B) $\frac{\partial f}{\partial x}$ is bounded and f is unbounded
- (C) $\frac{\partial f}{\partial x}$ is unbounded and f is bounded
- (D) $\frac{\partial f}{\partial x}$ and f are unbounded

Question 56 (JAM 2017 • Q5 • ID: JAM_2017_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $g(u, v) = f(u^2 - v^2)$, then

$$\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} =$$

- (A) $4(u^2 - v^2)f''(u^2 - v^2)$
- (B) $4(u^2 + v^2)f''(u^2 - v^2)$
- (C) $2f'(u^2 - v^2) + 4(u^2 - v^2)f''(u^2 - v^2)$
- (D) $2(u - v)^2 f''(u^2 - v^2)$

Question 57 (JAM 2017 • Q1 • ID: JAM_2017_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Consider the function $f(x, y) = 5 - 4 \sin x + y^2$ for $0 < x < 2\pi$ and $y \in \mathbb{R}$. The set of critical points of $f(x, y)$ consists of

- (A) a point of local maximum and a point of local minimum
- (B) a point of local maximum and a saddle point
- (C) a point of local maximum, a point of local minimum and a saddle point
- (D) a point of local minimum and a saddle point

Question 58 (JAM 2016 • Q60 • ID: JAM_2016_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 The maximum value of $f(x, y) = x^2 + 2y^2$ subject to the constraint $y - x^2 + 1 = 0$ is _____

END OF THE QUESTION PAPER

Question 59 (JAM 2016 • Q44 • ID: JAM_2016_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let f be a real valued function defined by

$$f(x, y) = 2 \ln \left(x^2 y^2 e^{\frac{y}{x}} \right), \quad x > 0, y > 0.$$

Then the value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ at any point (x, y) , where $x > 0, y > 0$, is _____

Question 60 (JAM 2016 • Q33 • ID: JAM_2016_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy}{|x|} & \text{if } x \neq 0 \\ 0 & \text{elsewhere} \end{cases}.$$

Then at the point $(0, 0)$, which of the following statement(s) is(are) TRUE ?

- (A) f is not continuous
- (B) f is continuous
- (C) f is differentiable
- (D) Both first order partial derivatives of f exist

Question 61 (JAM 2016 • Q18 • ID: JAM_2016_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 The function $f(x, y) = 3x^2y + 4y^3 - 3x^2 - 12y^2 + 1$ has a saddle point at

- (A) $(0, 0)$
- (B) $(0, 2)$
- (C) $(1, 1)$
- (D) $(-2, 1)$

Question 62 (JAM 2016 • Q17 • ID: JAM_2016_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy^2}{x+y} & \text{if } x+y \neq 0 \\ 0 & \text{if } x+y = 0 \end{cases}.$$

Then the value of $\left(\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y \partial x} \right)$ at the point $(0, 0)$ is

- (A) 0
- (B) 1
- (C) 2
- (D) 4

Question 63 (JAM 2015 • Q45 • ID: JAM_2015_Q45)

NAT

+1 M

Neg: Nil (0)

Q.5

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by

$$f(x, y, z) = \sin x + 2e^{\frac{y}{2}} + z^2$$

The maximum rate of change of f at $(\frac{\pi}{4}, 0, 1)$, correct upto three decimal places, is

Question 64 (JAM 2015 • Q44 • ID: JAM_2015_Q44)

NAT

+1 M

Neg: Nil (0)

Q.4

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \left(1 + \frac{x}{y}\right)^2, & y \neq 0 \\ 0, & y = 0 \end{cases}$$

If the directional derivative of f at $(0, 0)$ exists along the direction $\cos \alpha \hat{i} + \sin \alpha \hat{j}$, where $\sin \alpha \neq 0$, then the value of $\cot \alpha$ is

Question 65 (JAM 2015 • Q38 • ID: JAM_2015_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.8 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

At $(0, 0)$,

- (A) f is not continuous
- (B) f is continuous, and both f_x and f_y exist
- (C) f is differentiable
- (D) f_x and f_y exist but f is not differentiable

Question 66 (JAM 2015 • Q30 • ID: JAM_2015_Q30)

MCQ

+2 M

Neg: -0.67

Q.30

Let $f : \{ (x, y) \in \mathbb{R}^2 : x > 0, y > 0 \} \rightarrow \mathbb{R}$ be given by

$$f(x, y) = x^{\frac{1}{3}} y^{\frac{-4}{3}} \tan^{-1} \left(\frac{y}{x} \right) + \frac{1}{\sqrt{x^2 + y^2}}$$

Then the value of

$$g(x, y) = \frac{xf_x(x, y) + yf_y(x, y)}{f(x, y)}$$

- (A) changes with x but not with y
- (B) changes with y but not with x
- (C) changes with x and also with y
- (D) neither changes with x nor with y

SECTION - B**MULTIPLE SELECT QUESTIONS (MSQ)**

Q. 1 – Q. 10 carry two marks each.

Question 67 (JAM 2015 • Q25 • ID: JAM_2015_Q25)

MCQ

+2 M

Neg: -0.67

Q.25

For what real values of x and y , does the integral $\int_x^y (6 - t - t^2) dt$ attain its maximum?

- (A) $x = -3, y = 2$
- (B) $x = 2, y = 3$
- (C) $x = -2, y = 2$
- (D) $x = -3, y = 4$

Question 68 (JAM 2015 • Q15 • ID: JAM_2015_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 Suppose that the dependent variables z and w are functions of the independent variables x and y , defined by the equations $f(x, y, z, w) = 0$ and $g(x, y, z, w) = 0$, where $f_z g_w - f_w g_z = 1$. Which one of the following is correct?

(A) $z_x = f_w g_x - f_x g_w$

(B) $z_x = f_x g_w - f_w g_x$

(C) $z_x = f_z g_x - f_x g_z$

(D) $z_x = f_z g_w - f_z g_x$

Question 69 (JAM 2014 • Q25 • ID: JAM_2014_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} \frac{x}{|x|} \sqrt{x^2 + y^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$

Then $\frac{\partial f}{\partial x}(0, 0) + \frac{\partial f}{\partial y}(0, 0)$ equals

(A) -1

(B) 0

(C) 1

(D) 2

Question 70 (JAM 2014 • Q24 • ID: JAM_2014_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} x & \text{if } y = 0, \\ x - y^3 \sin(1/y) & \text{if } y \neq 0. \end{cases}$

Then at the point $(0, 0)$,

(A) f is NOT continuous(B) f is continuous but NOT differentiable(C) $\frac{\partial f}{\partial x}$ exists but $\frac{\partial f}{\partial y}$ does NOT exist(D) f is differentiable

Question 71 (JAM 2014 • Q5 • ID: JAM_2014_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 If $f(x, y, z) = x^2y + y^2z + z^2x$ for all $(x, y, z) \in \mathbb{R}^3$ and $\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$, then the value of $\nabla \cdot (\nabla \times \nabla f) + \nabla \cdot (\nabla f)$ at $(1, 1, 1)$ is

- (A) 0 (B) 3 (C) 6 (D) 9

Question 72 (JAM 2014 • Q3 • ID: JAM_2014_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $f(x, y) = \sum_{k=1}^{10} (x^2 - y^2)^k$ for all $(x, y) \in \mathbb{R}^2$. Then for all $(x, y) \in \mathbb{R}^2$,

- (A) $x \frac{\partial f}{\partial x}(x, y) - y \frac{\partial f}{\partial y}(x, y) = 0$ (B) $x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = 0$
 (C) $y \frac{\partial f}{\partial x}(x, y) - x \frac{\partial f}{\partial y}(x, y) = 0$ (D) $y \frac{\partial f}{\partial x}(x, y) + x \frac{\partial f}{\partial y}(x, y) = 0$

Question 73 (JAM 2012 • Q7 • ID: JAM_2012_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 For $c > 0$, if $a\hat{i} + b\hat{j} + c\hat{k}$ is the unit normal vector at $(1, 1, \sqrt{2})$ to the cone $z = \sqrt{x^2 + y^2}$, then

- (A) $a^2 + b^2 - c^2 = 0$ (B) $a^2 - b^2 + c^2 = 0$
 (C) $-a^2 + b^2 + c^2 = 0$ (D) $a^2 + b^2 + c^2 = 0$

Question 74 (JAM 2012 • Q3 • ID: JAM_2012_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases}$$

then

- (A) $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$
- (B) $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$
- (C) $f_x(0, 0) = 0$ and $f_y(0, 0) = 1$
- (D) $f_x(0, 0) = 1$ and $f_y(0, 0) = 1$

Question 75 (JAM 2009 • Q11 • ID: JAM_2009_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T(x, y, z) = xy^2 + 2z - x^2z^2$ be the temperature at the point (x, y, z) . The unit vector in the direction in which the temperature decreases most rapidly at $(1, 0, -1)$ is

- (A) $-\frac{1}{\sqrt{5}}\hat{i} + \frac{2}{\sqrt{5}}\hat{k}$
- (B) $\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{k}$
- (C) $\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$
- (D) $-\left(\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}\right)$

Question 76 (JAM 2007 • Q9 • ID: JAM_2007_Q9)

MCQ

+6 M

Neg: -2.0

9. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{x^2y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Which of the following statements holds regarding the continuity and the existence of partial derivatives of f at $(0, 0)$?

- (A) Both partial derivatives of f exist at $(0, 0)$ and f is continuous at $(0, 0)$
- (B) Both partial derivatives of f exist at $(0, 0)$ and f is NOT continuous at $(0, 0)$
- (C) One partial derivative of f does NOT exist at $(0, 0)$ and f is continuous at $(0, 0)$
- (D) One partial derivative of f does NOT exist at $(0, 0)$ and f is NOT continuous at $(0, 0)$

Question 77 (JAM 2006 • Q3 • ID: JAM_2006_Q3)

MCQ

+6 M

Neg: -2.0

3. Let $f(x, y) = x^5 y^2 \tan^{-1}\left(\frac{y}{x}\right)$. Then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ equals
- (A) $2f$
 (B) $3f$
 (C) $5f$
 (D) $7f$

Question 78 (JAM 2005 • Q11 • ID: JAM_2005_Q11)

MCQ

+6 M

Neg: -2.0

11. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be continuous and $g, h: \mathbf{R}^2 \rightarrow \mathbf{R}$ be differentiable. Let $F(u, v) = \int_v^u f(t) dt$, where $u = g(x, y)$ and $v = h(x, y)$. Then $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} =$
- (A) $f(g(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(h(x, y)) \left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \right]$
 (B) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$
 (C) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$
 (D) $f(g(x, y)) \left[\frac{\partial g}{\partial x} - \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of Two or Three Real Variables (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+2	0.24 to 0.26	27	MSQ	+2	A, D	53	NAT	+1	9.9 to 10.1
2	NAT	+1	3.0 to 3.0	28	MSQ	+2	B, D	54	MSQ	+2	A, D
3	MSQ	+2	A;C	29	MCQ	+2	C	55	MCQ	+2	B
4	MCQ	+2	D	30	NAT	+2	2 to 2	56	MCQ	+1	B
5	MCQ	+2	D	31	NAT	+1	2.70 to 2.72	57	MCQ	+1	D
6	NAT	+1	2.98 to 3.02	32	MCQ	+2	B	58	NAT	+2	2.0 to 2.0
7	MSQ	+2	B;D	33	MCQ	+2	D	59	NAT	+1	8.0 to 8.0
8	MSQ	+2	A;C;D	34	MCQ	+1	C	60	MSQ	+2	B;D
9	MCQ	+2	C	35	MCQ	+1	Marks to All	61	MCQ	+2	D
10	MSQ	+2	B;C	36	MCQ	+1	C	62	MCQ	+2	B
11	MCQ	+2	A	37	NAT	+2	5 to 5	63	NAT	+1	2.34 to 2.35
12	MCQ	+1	A	38	NAT	+1	2.6 to 2.6	64	NAT	+1	−1
13	NAT	+2	20.99 to 21.01	39	MSQ	+2	A, D	65	MSQ	+2	B;C
14	NAT	+1	0 to 0	40	MCQ	+2	B	66	MCQ	+2	D
15	MSQ	+2	A, B, C	41	MCQ	+2	B	67	MCQ	+2	A
16	MSQ	+2	A, B, C	42	MCQ	+2	A	68	MCQ	+2	A
17	MCQ	+2	B	43	MCQ	+1	B	69	MCQ	+2	C
18	MCQ	+2	D	44	MCQ	+1	B	70	MCQ	+2	D
19	MCQ	+1	B	45	NAT	+2	1140 to 1160	71	MCQ	+1	C
20	NAT	+2	5 to 5	46	NAT	+1	3 to 3	72	MCQ	+1	D
21	NAT	+2	18 to 18	47	NAT	+1	1 to 1	73	MCQ	+6	A
22	NAT	+1	14 to 14	48	NAT	+1	6.5 to 7.5	74	MCQ	+6	B
23	MSQ	+2	C,D	49	MCQ	+2	A	75	MCQ	+6	B
24	MCQ	+2	Marks to All	50	MCQ	+2	C	76	MCQ	+6	B
25	MCQ	+1	D	51	MCQ	+1	C	77	MCQ	+6	D
26	MCQ	+1	A	52	MCQ	+1	B	78	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.